

CHEMISTRY**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 9 printed pages.
4. It is divided into four sections and has twenty one questions in all.
5. Answer all questions.
6. Section A has fourteen subparts. Each question carries 1 mark.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B has ten questions. Each question carries 2 marks.
9. Section C has seven questions. Each question carries 3 marks.
10. Section D has three questions. Each question carries 5 marks.
11. Internal choices have been provided in two questions each in Sections B, C and D.
12. The intended marks for questions are given in brackets [].
13. All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.
14. Balanced equations must be given wherever possible and diagrams where they are helpful.
15. When solving numerical problems, all essential workings must be shown.
16. In working out problems, use the following data:
Gas constant $R = 1.987 \text{ cal deg}^{-1} \text{ mol}^{-1} = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$, $1 \text{ l atm} = 1 \text{ dm}^3 \text{ atm} = 101.3 \text{ J}$.
1 faraday = 96500 coulombs, Avogadro's number = 6.023×10^{23} .

Instruction to Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

SECTION A – 14 MARKS**Question 1**

- (A) Fill in the blanks by choosing the appropriate word(s) from those given in the brackets.
[benzyl alcohol, benzoic acid, Cannizzaro, Perkin's, 96500, 193000, one, two, coloured, colourless, paramagnetic, diamagnetic, salicylaldehyde, salicylic acid, Kolbe's, Reimer-Tiemann]
- (i) Benzaldehyde, which has no α -hydrogen atom, on heating with concentrated NaOH gives sodium benzoate and _____. The reaction is known as _____ reaction. (Recall) [1]

- (ii) The quantity of electricity needed to deposit one mole of copper from aqueous CuSO_4 is _____ coulombs, which is equal to _____ faraday(s). (*Apply*) [1]
- (iii) Zn^{2+} salts are _____ in aqueous solution, and Zn^{2+} is _____, because its 3d subshell is completely filled. (*Analyse*) [1]
- (iv) Phenol on treatment with chloroform and aqueous NaOH, followed by acidification, gives _____. This reaction is known as _____ reaction. (*Recall*) [1]
- (B) Select and write the correct alternative from the choices given below.
- (i) The correct order of elevation in boiling point of 0.1 m aqueous solutions of glucose, NaCl, BaCl_2 and $\text{K}_4[\text{Fe}(\text{CN})_6]$ (assume complete dissociation) is: (*Evaluate*) [1]
- (a) glucose > NaCl > BaCl_2 > $\text{K}_4[\text{Fe}(\text{CN})_6]$
- (b) $\text{K}_4[\text{Fe}(\text{CN})_6]$ > BaCl_2 > NaCl > glucose
- (c) BaCl_2 > $\text{K}_4[\text{Fe}(\text{CN})_6]$ > NaCl > glucose
- (d) NaCl > BaCl_2 > glucose > $\text{K}_4[\text{Fe}(\text{CN})_6]$
- (ii) Consider the following statements about glucose: [1]
- P: On prolonged heating with HI it forms n-hexane.
- Q: It reacts with three molecules of phenylhydrazine to form glucosazone.
- R: It gives Schiff's test and forms an addition product with NaHSO_3 .
- S: It is oxidised by bromine water to saccharic acid.
- Which of the statements are correct?
- (*Understand*)
- (a) P and Q only
- (b) P, Q and R only
- (c) Q and S only
- (d) P, R and S only
- (iii) One mole of the complex $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$, when treated with excess of AgNO_3 solution, precipitates two moles of AgCl. The correct formula of the complex is: (*Understand*) [1]
- (a) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
- (b) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
- (c) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
- (d) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$

- (iv) For the reaction $A + B \rightarrow C$ the following data were obtained: [1]

S. No.	[A] (mol L ⁻¹)	[B] (mol L ⁻¹)	Initial rate (mol L ⁻¹ s ⁻¹)
1	0.10	0.10	2.0×10^{-3}
2	0.20	0.10	8.0×10^{-3}
3	0.20	0.20	8.0×10^{-3}

The rate law for this reaction is:

(Apply)

- (a) rate = $k[A]^1[B]^1$
(b) rate = $k[A]^2[B]^0$
(c) rate = $k[A]^2[B]^1$
(d) rate = $k[A]^0[B]^2$
- (v) Which of the following gives a foul-smelling isocyanide when warmed with chloroform and alcoholic KOH? (Understand) [1]

- (a) $(CH_3)_2NH$
(b) $(CH_3)_3N$
(c) $C_6H_5NH_2$
(d) $C_6H_5NHCH_3$

- (vi) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: Anisole reacts with HI to give phenol and methyl iodide, and not iodobenzene and methanol.

Reason: The O–C₆H₅ bond has partial double-bond character due to resonance and is stronger than the O–CH₃ bond.

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.
- (vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: The molar conductivity of acetic acid increases steeply on dilution.

Reason: On dilution, the number of ions present per unit volume of the solution increases.

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.

- (C) Read the passage carefully and answer the questions that follow.

Phenol, C_6H_5OH , was first isolated from coal tar; today most of it is made from cumene (isopropylbenzene). Cumene is oxidised by air to cumene hydroperoxide, which is then decomposed by dilute acid. The process is economical because a second useful compound is obtained along with phenol.

The $-OH$ group activates the benzene ring towards electrophilic substitution, so phenol reacts with bromine water at room temperature, without any Lewis acid, to give a white precipitate. Phenol is also weakly acidic: it dissolves in aqueous $NaOH$, and electron-withdrawing groups such as $-NO_2$ at the ortho or para position make it a stronger acid. Phenol and its derivatives are used to make dyes, plastics and drugs such as aspirin.

- (i) Name the compound obtained along with phenol in the cumene process. (*Recall*) [1]
- (ii) Name the white precipitate formed when phenol reacts with bromine water, and write the equation. (*Understand*) [1]
- (iii) Why is p-nitrophenol a stronger acid than phenol? (*Evaluate*) [1]

SECTION B – 20 MARKS

Question 2

The standard reduction potentials of some electrodes at 298 K are:

Electrode	E°
Al^{3+}/Al	-1.66 V
Zn^{2+}/Zn	-0.76 V
Ni^{2+}/Ni	-0.25 V
Cu^{2+}/Cu	$+0.34\text{ V}$
Ag^+/Ag	$+0.80\text{ V}$

(*Apply*)

- (i) Arrange the metals in decreasing order of their reducing power. [1]
- (ii) Which of these metals cannot liberate hydrogen from dilute HCl ? Justify. [1]

Question 3

An alkene [A], C_4H_8 , on ozonolysis followed by treatment with zinc and water gives two compounds [B] and [C]. Compound [B] reduces Tollens' reagent but does not give the iodoform test. Compound [C] does not reduce Tollens' reagent but gives a yellow precipitate with iodine and sodium hydroxide. Identify [A], [B] and [C] and write the iodoform reaction of [C]. (*Understand*) [2]

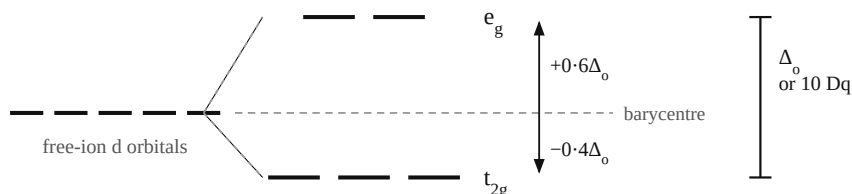
Question 4

- (i) Two students were asked to describe the complex ion $[CoF_6]^{3-}$. Priya said that it is an inner-orbital complex with d^2sp^3 hybridisation and is diamagnetic. Arjun said that it is an outer-orbital complex with sp^3d^2 hybridisation and is paramagnetic with four unpaired electrons. (Atomic number of $Co = 27$)
- (a) Whose description is correct? (*Evaluate*) [1]

- (b) Justify your answer using the electronic configuration of the central ion. (*Analyse*) [1]

OR

- (ii) Observe the diagram below of splitting of d orbitals in an octahedral field and answer the questions that follow.



- (a) Write the electronic configuration of a d^6 ion if $\Delta_o > P$. (*Understand*) [1]
 (b) How many unpaired electrons does a d^6 ion have if $\Delta_o < P$? Comment on its magnetic nature. (*Analyse*) [1]

Question 5

Phosphorus-32, used in the treatment of leukaemia, has a half-life of 14.3 days. A patient is given a dose containing 10 mg of P-32. What mass of P-32 remains in the body after 57.2 days? Assume that none of it is eliminated by biological processes. (*Apply*) [2]

Question 6

How can the following conversions be carried out? (Write the chemical equations.)

- (i) Aniline to benzene (*Recall*) [1]
 (ii) Chlorobenzene to biphenyl (*Recall*) [1]

Question 7

Substantiate the following statements with reasons.

- (i) Mn^{2+} compounds are more stable than Fe^{2+} compounds towards oxidation to the +3 state. (Atomic number of Mn = 25 and Fe = 26) (*Understand*) [1]
 (ii) Zn, Cd and Hg are not regarded as transition elements. (Atomic number of Zn = 30, Cd = 48 and Hg = 80) (*Understand*) [1]

Question 8

Write the Nernst equation and calculate the emf of the following cell at 298 K: [2]



Given: $E^\circ(Mg^{2+}/Mg) = -2.37 V$, $E^\circ(Ag^+/Ag) = +0.80 V$ (*Apply*)

Question 9

- (i) Illustrate the following reactions with an example in each case:
 (a) Hofmann bromamide degradation (*Recall*) [1]

- (b) Gattermann reaction (*Recall*) [1]

OR

- (ii) An aromatic compound [A], C_7H_7NO , on heating with bromine and KOH gives [B], C_6H_7N . [B] reacts with $NaNO_2$ and dilute HCl at 273–278 K to give [C]. [C] on warming with water gives [D], which gives a violet colour with neutral $FeCl_3$ solution. Write the structures of [A], [B], [C] and [D]. (*Understand*) [2]

Question 10

- The rate of a reaction doubles when the temperature is raised from 300 K to 310 K. Calculate the activation energy of the reaction. (*Apply*) [2]

Question 11

- A solution of 0.60 g of KCl in 400 g of water boils at 373.0206 K at 1 atm. Calculate the observed molar mass of KCl and its van't Hoff factor, and comment on the result. (Given: K_b of water = $0.52 \text{ K kg mol}^{-1}$, boiling point of pure water = 373 K, $K = 39$, $Cl = 35.5$) (*Analyse*) [2]

SECTION C – 21 MARKS

Question 12

Write balanced chemical equations for the following and name the organic product formed in each case.

- (i) Ethanol is heated with thionyl chloride in the presence of pyridine. Why is this method preferred for preparing alkyl chlorides? (*Understand*) [1]
- (ii) Bromoethane is heated with sodium iodide in dry acetone. (*Recall*) [1]
- (iii) Chloroform is exposed to air and sunlight. (*Understand*) [1]

Question 13

- (i) A first-order reaction is 30% complete in 12 minutes. Calculate the time required for it to be 75% complete. (*Apply*) [2]
- (ii) The rate law of a reaction is $\text{rate} = k[A]^{1/2}[B]$. Write its overall order and the unit of its rate constant. (*Understand*) [1]

Question 14

- (i) Three compounds [X], [Y] and [Z] have the molecular formula C_3H_8O . [X] and [Y] liberate hydrogen with sodium metal but [Z] does not. [Y] gives a yellow precipitate on warming with iodine and NaOH, but [X] does not.
- (a) Identify [X], [Y] and [Z]. (*Understand*) [1]
- (b) Which of the three has the lowest boiling point? Give a reason. (*Understand*) [1]
- (ii) Write the chemical equation for the reaction of [Y] with iodine and sodium hydroxide. (*Apply*) [1]

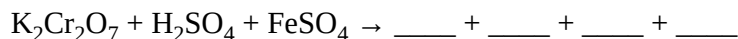
OR

- (iii) How will the following be converted? (Give chemical equations.)
- (a) Phenol to anisole (*Recall*) [1]
- (b) Methanal to ethanol (*Recall*) [1]

- (c) Chlorobenzene to phenol (*Recall*) [1]

Question 15

- (i) Complete and balance the following chemical equation: [1]



(*Understand*)

- (ii) Zirconium ($Z = 40$) and hafnium ($Z = 72$) have almost identical atomic radii. Explain. [1]

(*Analyse*)

- (iii) What is observed when acidified KMnO_4 solution is added to a warm solution of oxalic acid? [1]

Write the ionic equation. (*Apply*)

Question 16

- (i) What are the products of hydrolysis of lactose? Name the linkage between them. (*Recall*) [1]

- (ii) What happens to the secondary and tertiary structures of the protein when an egg is boiled? [1]

What is this change called? (*Understand*)

- (iii) Name the vitamins whose deficiency causes (a) scurvy (b) beriberi. (*Recall*) [1]

Question 17

- (i) Two electrolytic cells containing solutions of CuSO_4 and AgNO_3 are connected in series. A steady current of 2.0 A is passed through them for 40 minutes. (Atomic mass of Cu = 63.5, Ag = 108)

- (a) How much charge passes through each cell? (*Apply*) [1]

- (b) Calculate the masses of copper and silver deposited at the respective cathodes. (*Analyse*) [2]

OR

- (ii) The limiting molar conductivities of HCl, CH_3COONa and NaCl are 426.2, 91.0 and 126.5 $\text{S cm}^2 \text{mol}^{-1}$ respectively at 298 K.

- (a) Using Kohlrausch's law, calculate the limiting molar conductivity of acetic acid. (*Apply*) [2]

- (b) If the molar conductivity of an acetic acid solution at the same temperature is 39.05 $\text{S cm}^2 \text{mol}^{-1}$, calculate its degree of dissociation. (*Apply*) [1]

Question 18

Write the chemical equations for the following conversions: (*Recall*)

- (i) Ethyl isocyanide to N-methylethanamine [1]

- (ii) Aniline to iodobenzene [1]

- (iii) Methyl chloride to ethylamine [1]

SECTION D – 15 MARKS

Question 19

- (i) An organic compound [A], C_3H_6O , does not reduce Fehling's solution but gives a yellow precipitate on warming with iodine and sodium hydroxide. [A] on heating with zinc amalgam and concentrated HCl gives [B]. When [A] is treated with dilute $Ba(OH)_2$ it undergoes self-condensation to give [C], $C_6H_{12}O_2$, which on heating loses a molecule of water to give [D], $C_6H_{10}O$.
- (a) Identify the compounds [A], [B], [C] and [D]. (*Understand*) [2]
- (b) Write the reaction for the formation of [B] from [A]. What is the reaction called? (*Recall*) [1]
- (c) Give a chemical test to distinguish [A] from propanal. (*Analyse*) [1]
- (d) Arrange methanal, propanal and propanone in increasing order of reactivity towards nucleophilic addition. Give a reason. (*Evaluate*) [1]

OR

- (ii) Identify the compounds [A], [B] and [C] in the following reactions: (*Analyse*)
- (a)
$$C_6H_5COOH \xrightarrow{PCl_5} [A] \xrightarrow{H_2, Pd/BaSO_4} [B] \xrightarrow{conc. NaOH} [C] + C_6H_5COONa$$
 [1½]
- (b)
$$CH_3CH_2Br \xrightarrow{KCN (alc.)} [A] \xrightarrow[(ii) H_2O]{(i) SnCl_2/HCl} [B] \xrightarrow{dil. NaOH} [C]$$
 [1½]
- (iii) Write chemical equations to convert the following: (*Recall*)
- (a) Ethanoic acid to propanone [1]
- (b) Toluene to benzaldehyde [1]

Question 20

- (i) The complex ion $[Co(NH_3)_4Cl_2]^+$ exists in two geometrical isomeric forms.
- (a) Write the IUPAC name of the complex ion. (*Recall*) [1]
- (b) Draw the structures of the two isomers. (*Understand*) [1]
- (ii) What type of structural isomers are $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5(SO_4)]Br$? How can they be distinguished by a chemical test? (*Analyse*) [2]
- (iii) A complex of composition $CoCl_3 \cdot 5NH_3$ gives two moles of AgCl per mole of complex when treated with excess $AgNO_3$ solution. Write its structural formula and IUPAC name. (*Evaluate*) [1]

Question 21

- (i) Answer the following:
- (a) 250 mL of an aqueous solution contains 2.4 g of a polymer. The osmotic pressure of the solution at 27 °C is 4.5×10^{-3} atm. Calculate the molar mass of the polymer. (*Apply*) [2]
- (b) Benzoic acid (C_6H_5COOH) dimerises in benzene. A solution of 2.44 g of benzoic acid in 50 g of benzene lowers the freezing point of benzene by 1.12 K. Calculate the degree of association of benzoic acid. [2]
- (K_f for benzene = $5.12 \text{ K kg mol}^{-1}$, molar mass of benzoic acid = 122 g mol^{-1}) (*Apply*)

- (c) The molar masses of calcium nitrate and urea are determined by the osmotic pressure method. [1]
How will their observed molar masses compare with their theoretical values? Justify your answer. (*Evaluate*)

OR

- (ii)
- (a) Arrange the following aqueous solutions in increasing order of their boiling points. Give [2]
reasons in support of your answer.
(1) 9.0 g of glucose per litre (molar mass = 180 g mol^{-1})
(2) 6.0 g of urea per litre (molar mass = 60 g mol^{-1})
(3) 5.85 g of sodium chloride per litre (molar mass = 58.5 g mol^{-1})
(4) 11.1 g of calcium chloride per litre (molar mass = 111 g mol^{-1}) (*Analyse*)
- (b) Calculate the molality and the mole fraction of ethylene glycol in a solution made by [2]
dissolving 31 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) in 500 g of water. (C = 12, H = 1, O = 16) (*Apply*)
- (c) A mixture of ethanol and water shows positive deviation from Raoult's law. What type of [1]
azeotrope does it form? Justify your answer in terms of the intermolecular interactions in the solution. (*Evaluate*)

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